WATER SUPPLY APP

Scenario Analysis

Operational scenarios for a fully automated water saving system — normal, degraded and edge-case behavior

Design document — Backend automation layer
August 2026

1. Purpose

This document defines the scenarios the Water Supply App's automated backend must handle once create and delete operations are no longer performed by end users, but by the system itself — through a scheduler, IoT sensors, and a rules engine, as described in the automation feasibility analysis.

Each scenario specifies a trigger condition, the automated response the system executes internally, and the signal made visible to the user (dashboard state, alert, or notification). The goal is to validate, before implementation, that every state the system can reach in production has a defined and predictable behavior — including failure modes such as a missing sensor reading or an out-of-range water quality test.

This scenario catalog complements the backend architecture document: it does not redefine the data model or the API contracts, but specifies how the KPI engine, the irrigation scheduler and the alerting module should behave under each condition.

2. Method

Scenarios are grouped into three categories, consistent with how a 100% automated system reaches its states:

- Normal operation — the system behaves as expected, no manual intervention required.
- Degraded operation — a resource or a data source is under stress or unavailable, but the system continues operating with a defined fallback.
- Edge cases — infrequent but foreseeable situations (prolonged drought, planned maintenance, simultaneous demand peaks) that require a specific, pre-defined handling rule rather than improvised behavior.

For each scenario, the same four-column structure is used: the scenario name, the trigger condition that starts it, the automated response executed by the backend (scheduler, rules engine, or sensor-driven update), and the signal visible to the end user. This structure keeps the analysis directly actionable for implementation and for QA test-case design.

3. Scenario catalog — summary

Scenario	Trigger	Automated system response	User-visible signal
Normal irrigation cycle	Soil humidity below seasonal threshold	Scheduler auto-triggers irrigation on the affected zone; KPI engine logs planned vs. actual volume	Zone status changes to “In progress”, then “Completed”
Irrigation not needed	Soil humidity above seasonal threshold	Scheduler skips the cycle; no water drawn	Zone status “Postponed” with reason displayed
Reservoir critical	Tank level < 20%	New irrigations on non-priority zones are queued, not started	Red alert on dashboard + reservoir card
Sensor failure / no data	No reading received for > 15 min from a source	System falls back to last known value; marks data as stale	“Data unavailable” badge on the affected metric
Water quality out of range	pH or turbidity outside compliance thresholds	Source flagged non-compliant; irrigation from that source paused	Red “Non-compliant” tag + notification
Rainwater near overflow	Rain tank level > 90%	System recommends prioritizing rainwater consumption	Orange “Near overflow” tag on rain tank card
Abnormal drip flow	Measured flow deviates > 25% from theoretical	Zone flagged for inspection; excluded from auto-scheduling until resolved	Red “Abnormal flow” alert on the zone

Scenario	Trigger	Automated system response	User-visible signal
Prolonged drought	Reservoir trend declining over several consecutive days	Risk level escalates (Normal → Vigilance → Alert → Critical)	Risk gauge updates; recommendations displayed
Planned maintenance	A zone or source is marked under maintenance	Zone excluded from the scheduler for the maintenance window	Zone shown as “Under maintenance”, greyed out
Simultaneous demand peak	Multiple zones eligible to irrigate at the same time	Scheduler prioritizes by criticality / lowest recent coverage	Queue order shown in the irrigation plan

Table 1 — Summary of the ten scenarios covered by this analysis

4. Detailed scenario sheets

The four scenarios below are detailed further, as they involve the most cross-module coordination (irrigation, KPI engine, alerting) and therefore carry the highest implementation risk.

4.1 Reservoir critical level

PRECONDITION

The system continuously tracks the reservoir level via the WaterSource entity, updated by sensor readings or scheduled polling.

TRIGGER

Reservoir level drops below the configured critical threshold (default 20%, adjustable per zone/season).

AUTOMATED RESPONSE — STEP BY STEP

- The monitoring module detects the threshold breach at the next polling cycle.
- The irrigation scheduler re-evaluates the day's plan and queues (does not cancel) irrigations for non-priority zones.
- Priority zones (defined by crop criticality) may still proceed, drawing down the reservoir further only if justified.
- An alert is generated and pushed to the notifications module.
- The KPI engine logs the event so the water-saved calculation reflects the postponement.

EXPECTED OUTCOME

No irrigation is silently lost — it is queued and resumed automatically once the reservoir recovers above threshold, or explicitly overridden by a supervisor.

RELATED MODULE / ENDPOINT

GET /api/dashboard/water-savings, monitoring & alerts module (Objective 5), IrrigationSchedule entity.

4.2 Sensor failure or missing data

PRECONDITION

A given source (tank level sensor, flow meter, quality probe) is expected to report at a fixed interval.

TRIGGER

No new reading is received for more than 15 minutes (configurable per sensor type).

AUTOMATED RESPONSE — STEP BY STEP

- The ingestion layer flags the metric as “stale” rather than raising a hard error.
- Downstream calculations (KPI, scheduler) fall back to the last known valid value, tagged as an estimate.

- If the gap exceeds a second, longer threshold (e.g. 2 hours), the system escalates to a technical alert distinct from an operational one.
- No automatic irrigation decision is based solely on stale data beyond the fallback window.

EXPECTED OUTCOME

The dashboard never silently displays wrong data as if it were live — it is explicitly marked as unavailable or estimated, per the non-functional requirement on tolerance to missing data.

RELATED MODULE / ENDPOINT

WaterConsumption / WaterSource ingestion services, Task 5.5 non-functional constraints (data completeness indicator).

4.3 Water quality out of range

PRECONDITION

Water quality tests (pH, turbidity) are recorded per source, either by a probe or a manual entry, and compared against compliance thresholds.

TRIGGER

A new test result falls outside the compliant range for its source.

AUTOMATED RESPONSE — STEP BY STEP

- The source is immediately flagged as non-compliant in the WaterQualityTest / WaterSource relation.
- The irrigation scheduler excludes that source from new cycles until a compliant retest is logged.
- A red alert is raised, surfaced at the top of the Water Quality module and on the dashboard alert count.
- If the source has no compliant alternative, the affected zones are marked “Postponed — quality hold” rather than proceeding.

EXPECTED OUTCOME

No irrigation draws from a source known to be non-compliant; the block is lifted automatically as soon as a passing retest is recorded.

RELATED MODULE / ENDPOINT

Water Quality Testing Logs module, monitoring & alerts module (Objective 5).

4.4 Prolonged drought / escalating risk

PRECONDITION

The Drought Alert System tracks reservoir trend and season over a rolling multi-day window, not just the instantaneous level.

TRIGGER

Reservoir level shows a sustained downward trend across several consecutive days without sufficient rainwater recovery.

AUTOMATED RESPONSE — STEP BY STEP

- The risk engine recalculates the risk level daily: Normal → Vigilance → Alert → Critical.
- Each escalation step updates the recommendation panel (e.g. reduce non-essential washing, prioritize rainwater).
- At “Critical”, the scheduler proposes automatic postponement of irrigation for the lowest-priority zones.
- All level changes are timestamped and kept in the risk history for later review.

EXPECTED OUTCOME

The team is never surprised by a sudden critical shortage — the risk level rises progressively, with recommendations that scale with severity.

RELATED MODULE / ENDPOINT

Drought Alert System module; feeds back into Objective 6 KPI (a postponement to preserve reserves still counts toward long-term water savings).

5. Conclusion

These ten scenarios cover the operational range the automated Water Supply App is expected to handle without manual create/delete actions: routine irrigation cycles, resource stress conditions, data-availability issues, and rare but foreseeable edge cases. Together with the backend architecture document, this scenario catalog gives the development team a concrete basis for implementing the scheduler, the rules engine, and the alerting logic, and gives QA a direct source of test cases before the first release.

As implementation progresses, new scenarios uncovered during testing should be added to this catalog using the same four-column structure, so the document remains the single reference for expected system behavior under any condition.